

The background features decorative blue geometric shapes in the corners. On the top-left, there are several overlapping parallelograms in different shades of blue. On the top-right, a single parallelogram is visible. On the bottom-left, a parallelogram is partially shown. On the bottom-right, there is a larger, more complex arrangement of overlapping parallelograms in various shades of blue.

Data presentation

The collection of numerical information often leads to large masses of data which, if they are to be **understood, or presented effectively**, must be summarised and analysed in some way.

Various methods are seen in presentation of data but the most common are **tabular and graphical methods**.

1.THE TABULATION OF DATA

The presentation of data in form of table is called **tabulation**.

A.Simple table: Tables that express only a **single feature** of a particular data are called simple tables.

Simple tables –

Data relating to only **one** characteristics

<u>Gender</u>	No of students
Boys	9
Girls	29

B. Complex table: Tables that represent **multiple features** of data, such as the male and female population of a particular group and the total population.

Double table -

Data relating to only **2** characteristics

<u>Gender</u>	<u>Food habit</u>	
	Vegetarians	Non Vegetarians
Boys	2	7
Girls	5	24

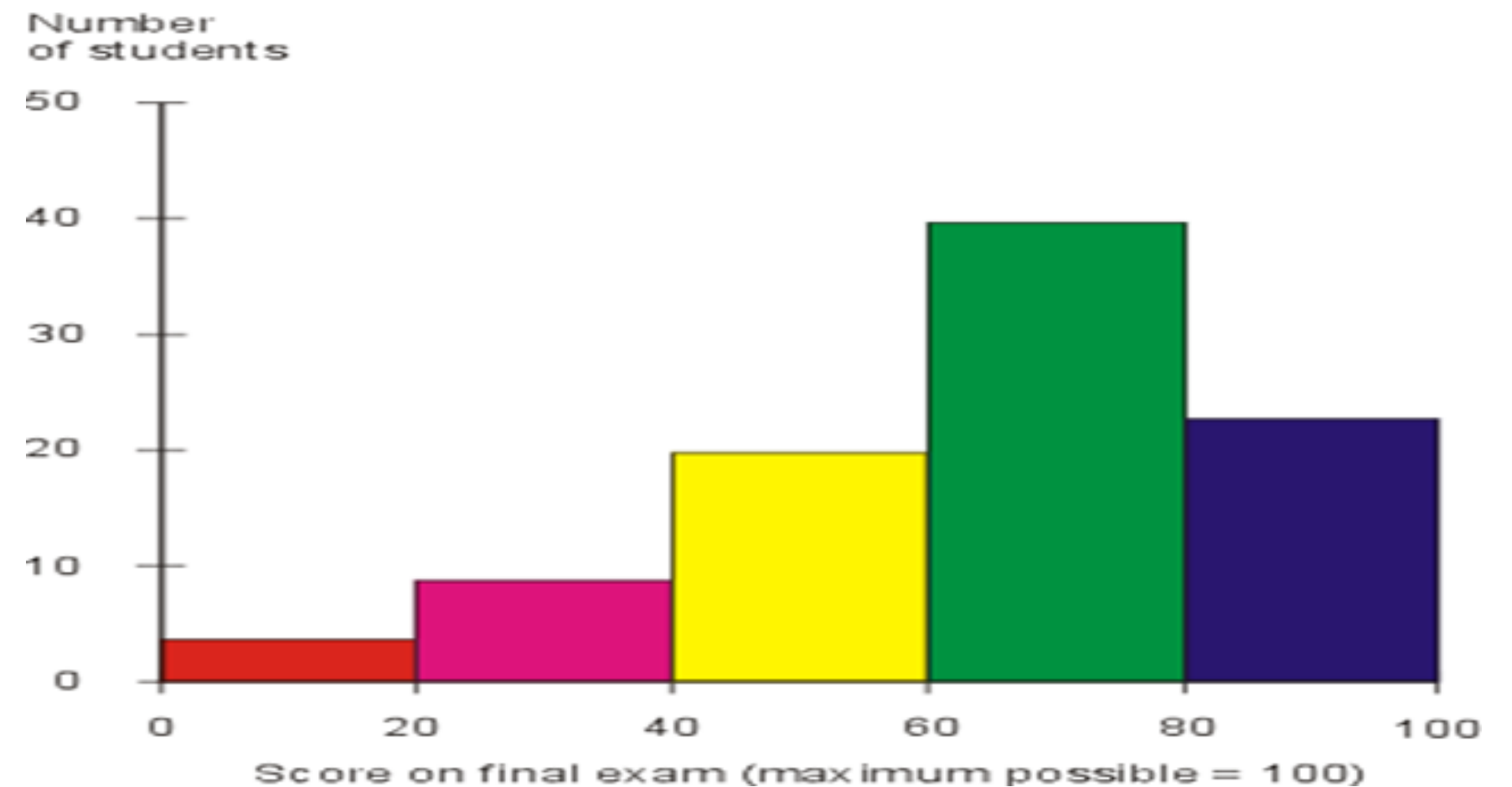
2-THE GRAPHICAL REPRESENTATION OF DATA

Graphical representation is the visual display of data using plots and diagrams or charts.

There are different types of graphical representation and which is used depends on the nature of the data and the nature of the statistical results.

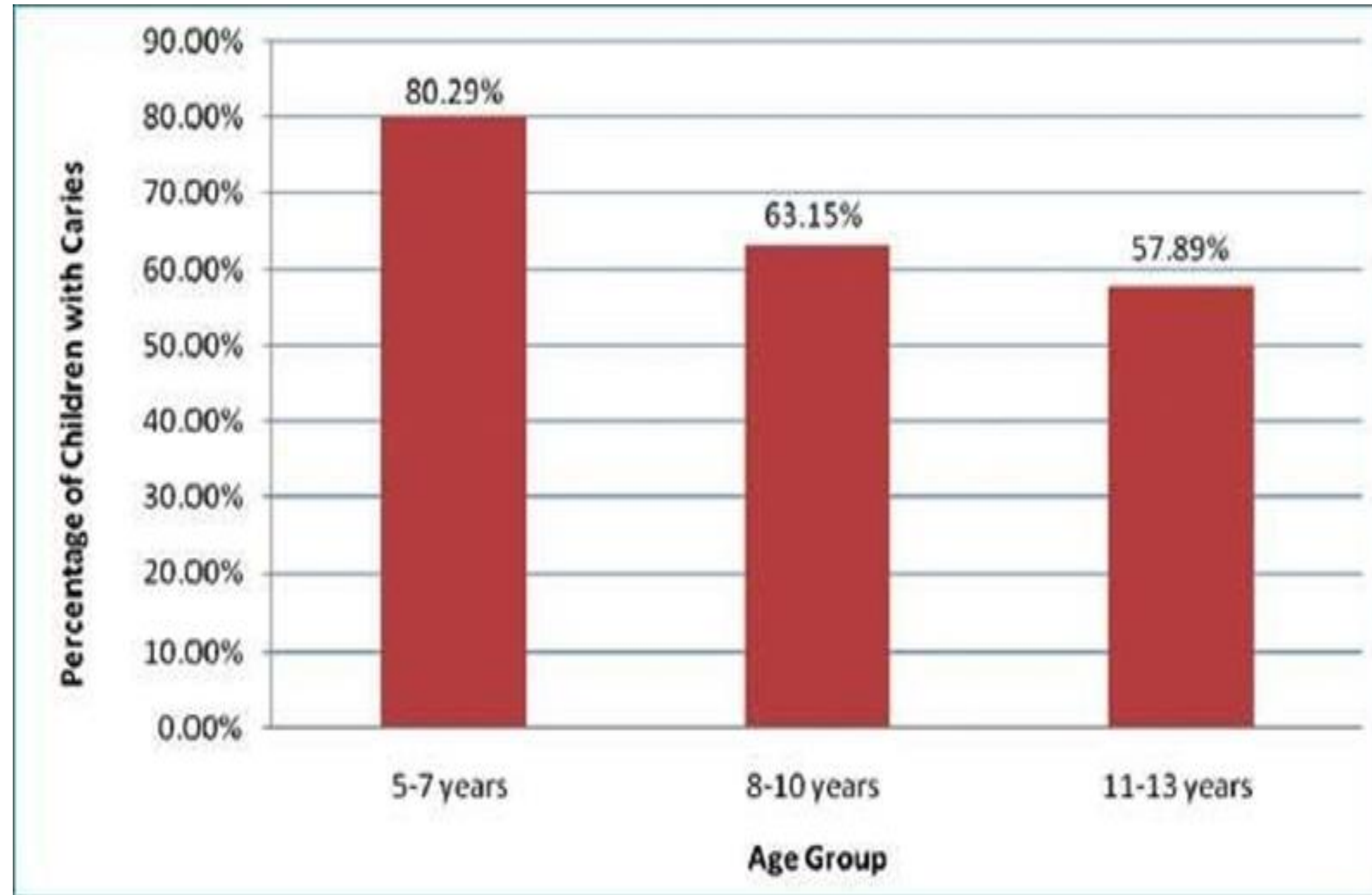
(a) Histogram

A “histogram” is a diagram consists of a collection of rectangles whose **height** represents the class frequency and whose extent represents the **class width**. This type of graph is used with **quantitative data**.

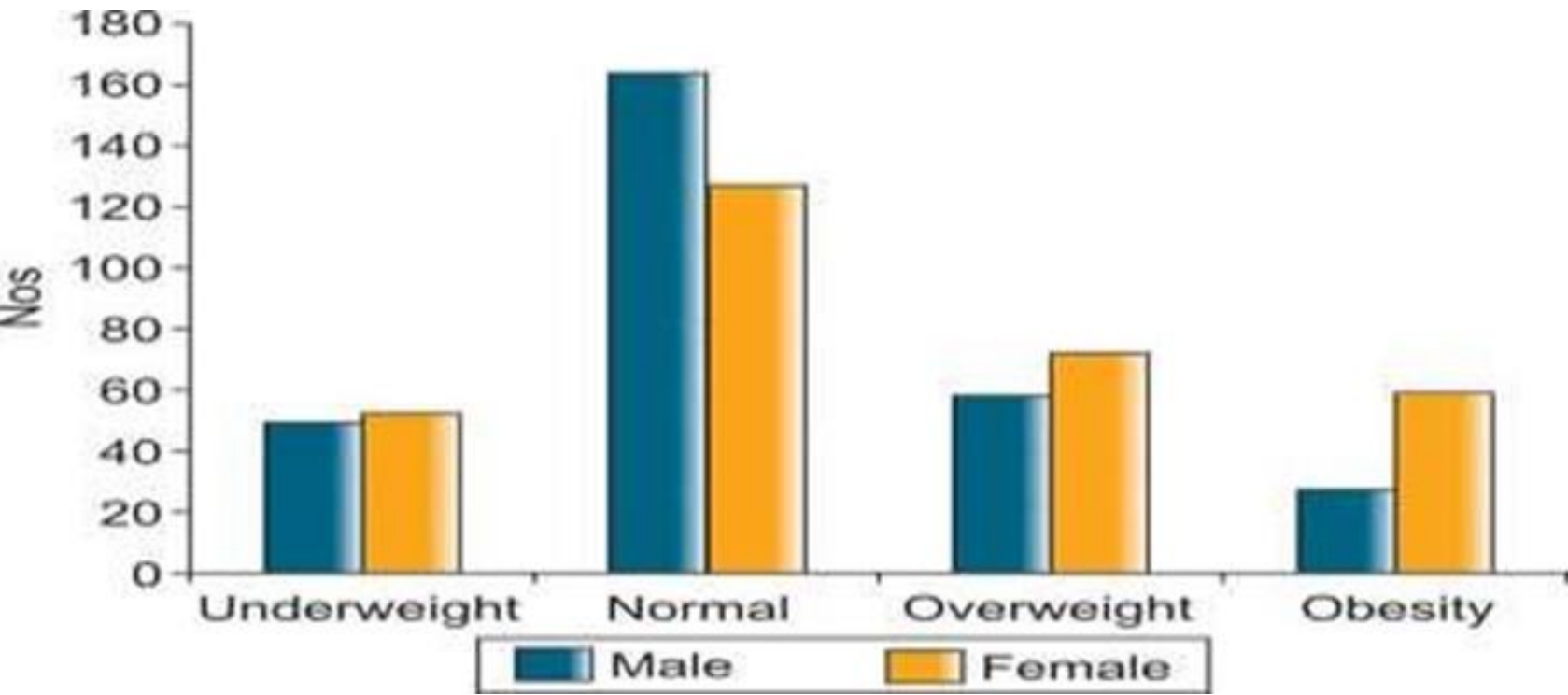


b- Bar graph: the variable is plotted in the form of bars. the height of the bar is equal to the **frequencies or percentage**. It is used for **qualitative** type of data .

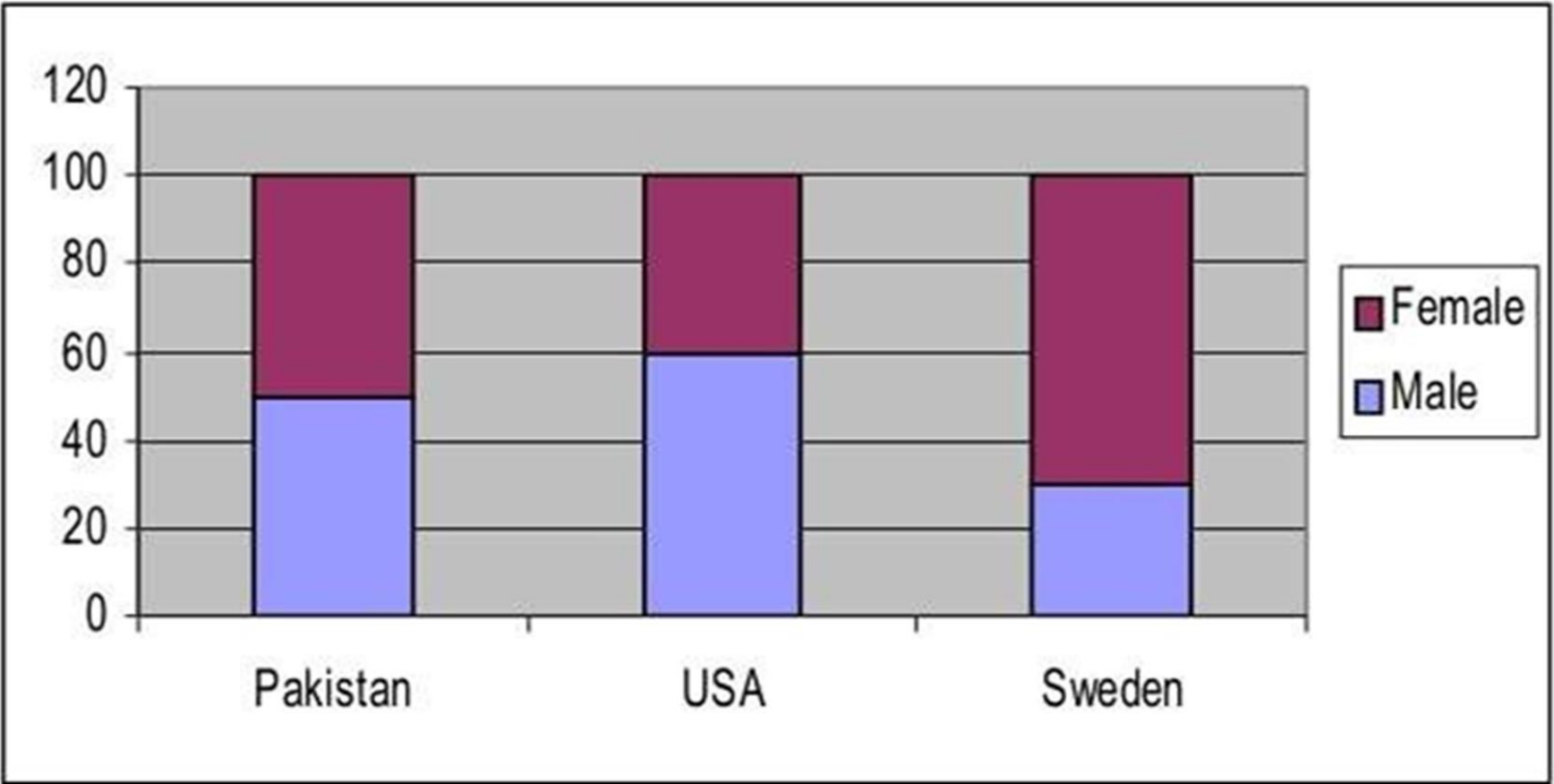
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2-Multiple bar graph: used for data which are made of two or more components, different color is given for each component.

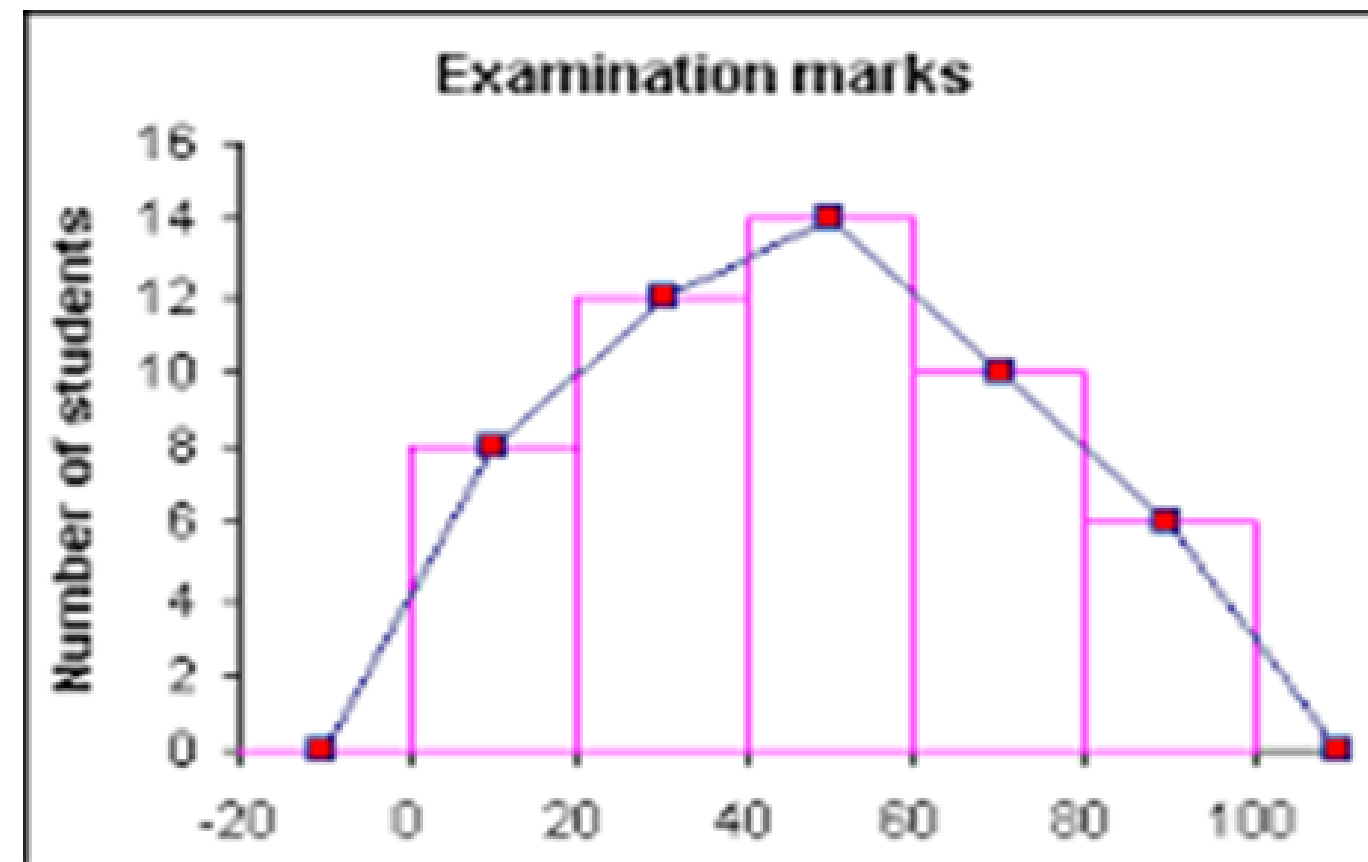
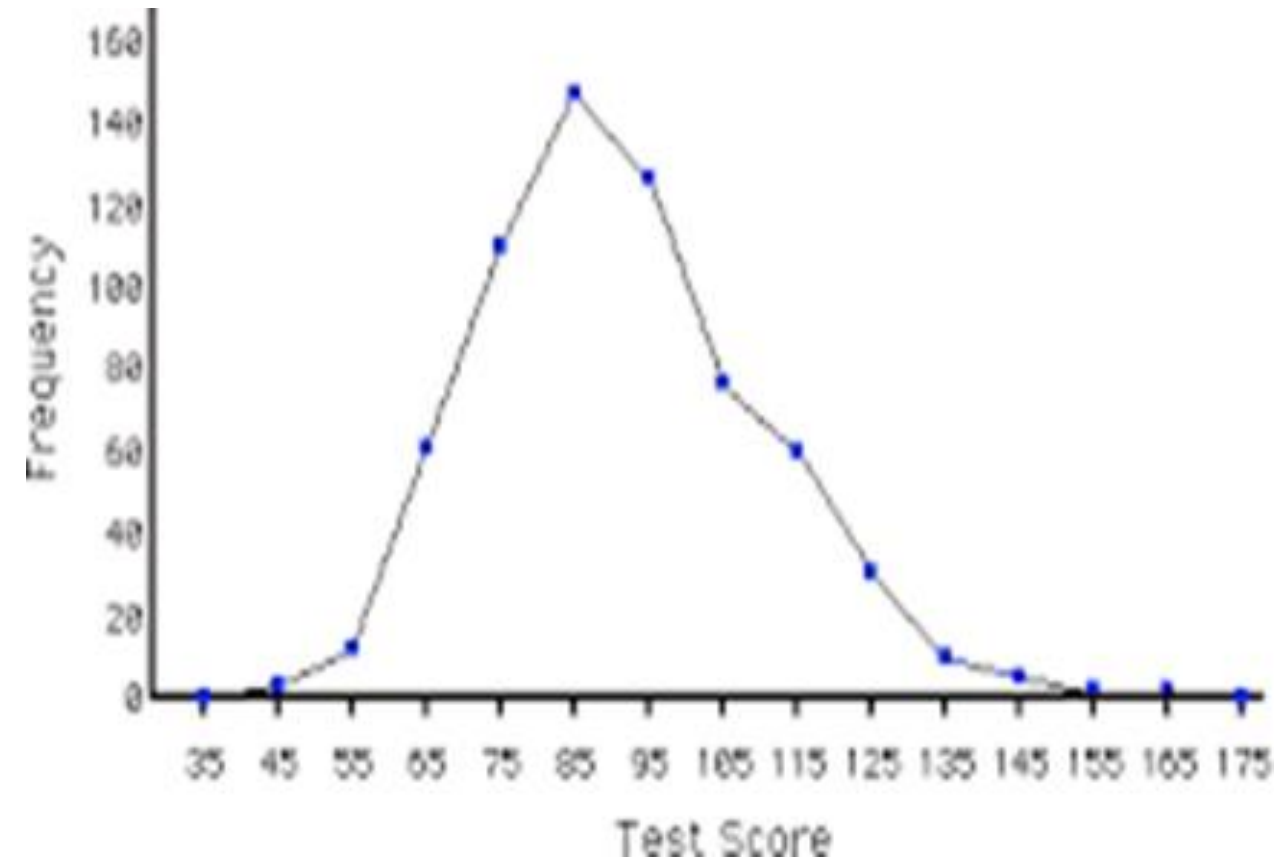


3-Sub-divided bar graph (component bar): When there are many categories and they have further subcategories, the bars may be divided into parts, each part representing a certain item. The components are shown with different colors.



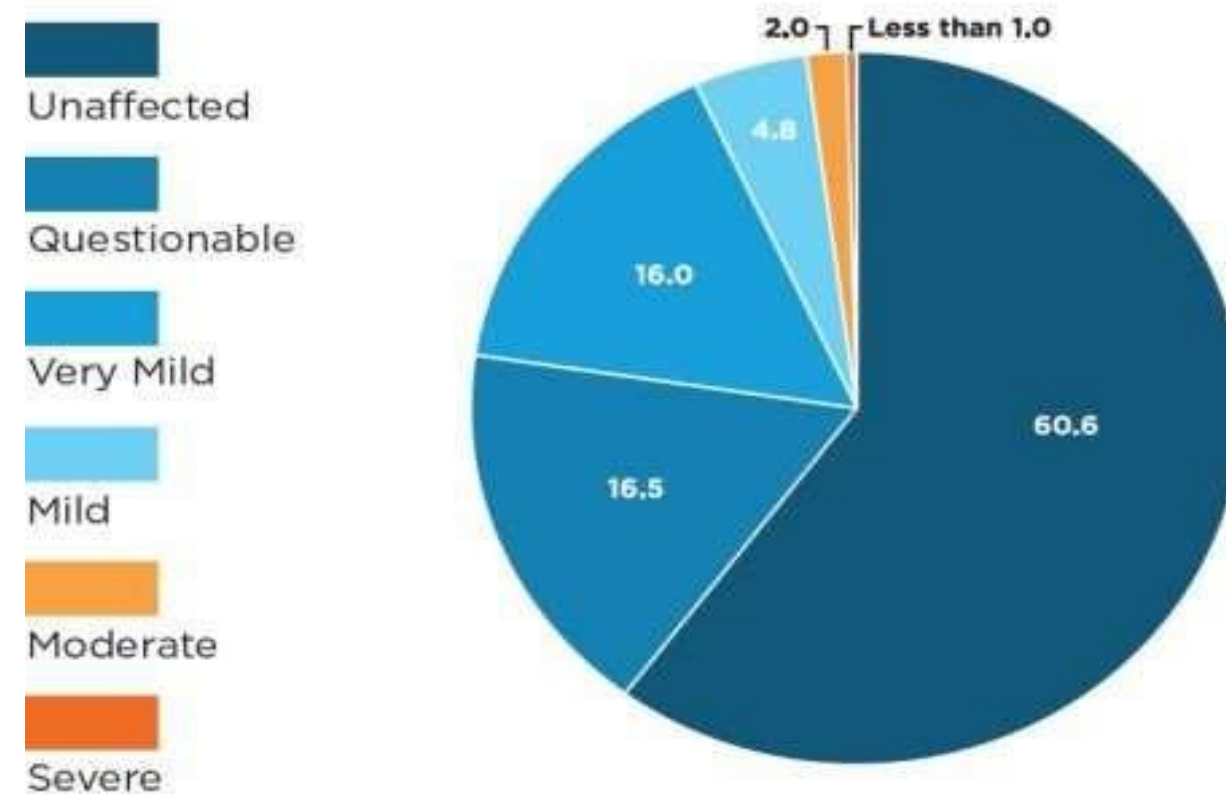
c-The Frequency Polygon

Using the fact that each class represented by its class mid-point, we may plot the class mid-points against the class frequencies to obtain a display of single points so it is an area diagram of frequency distribution over a histogram, obtained by joining the mid points of the histogram block.



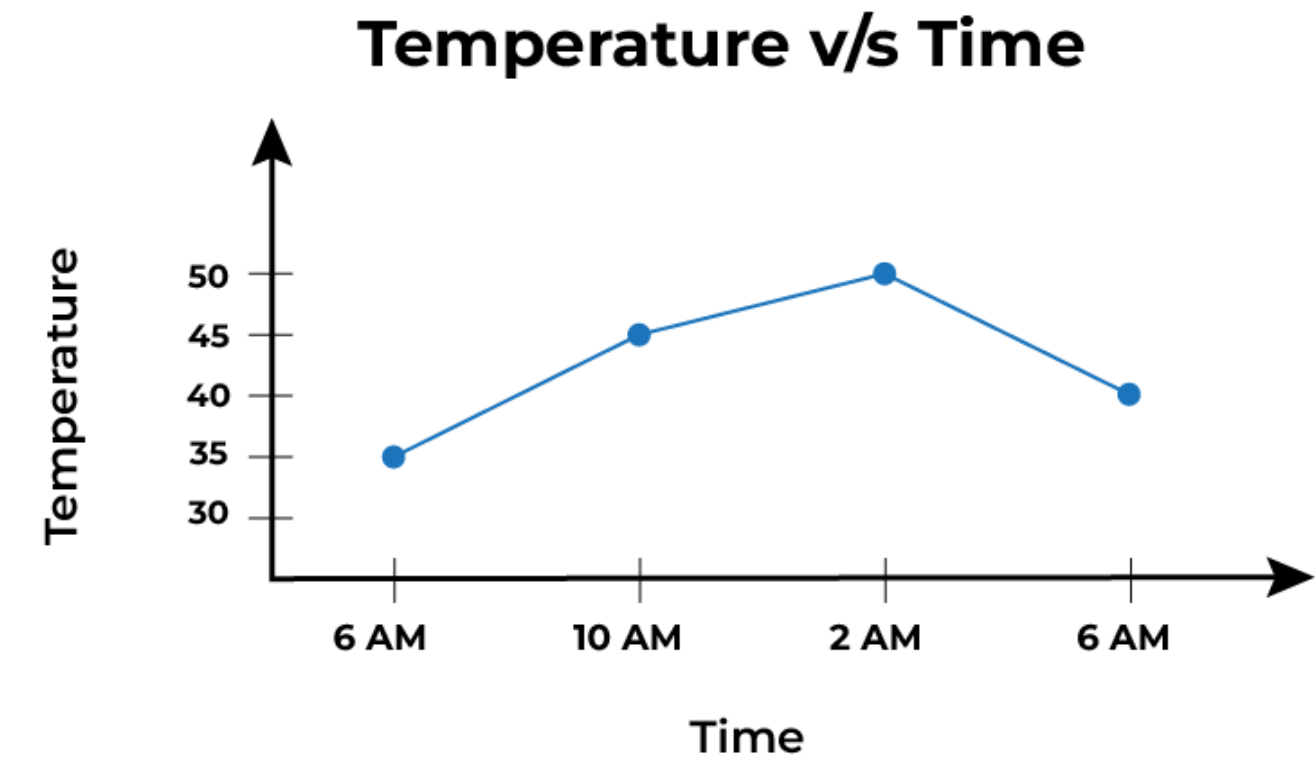
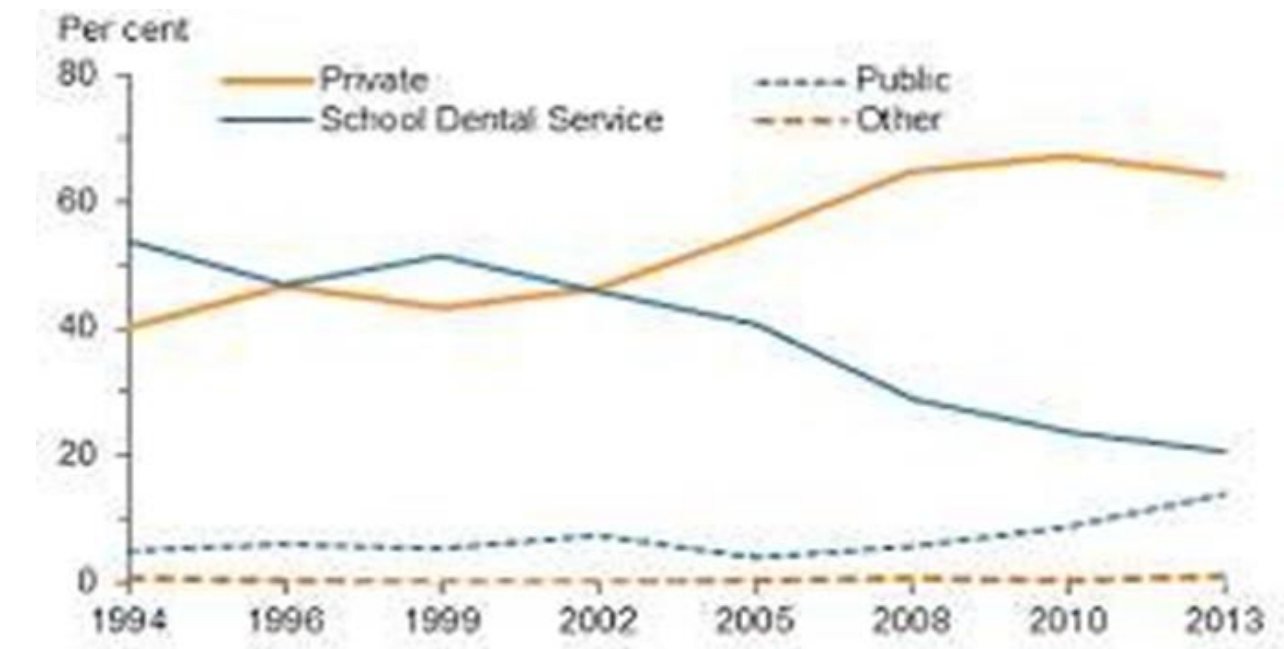
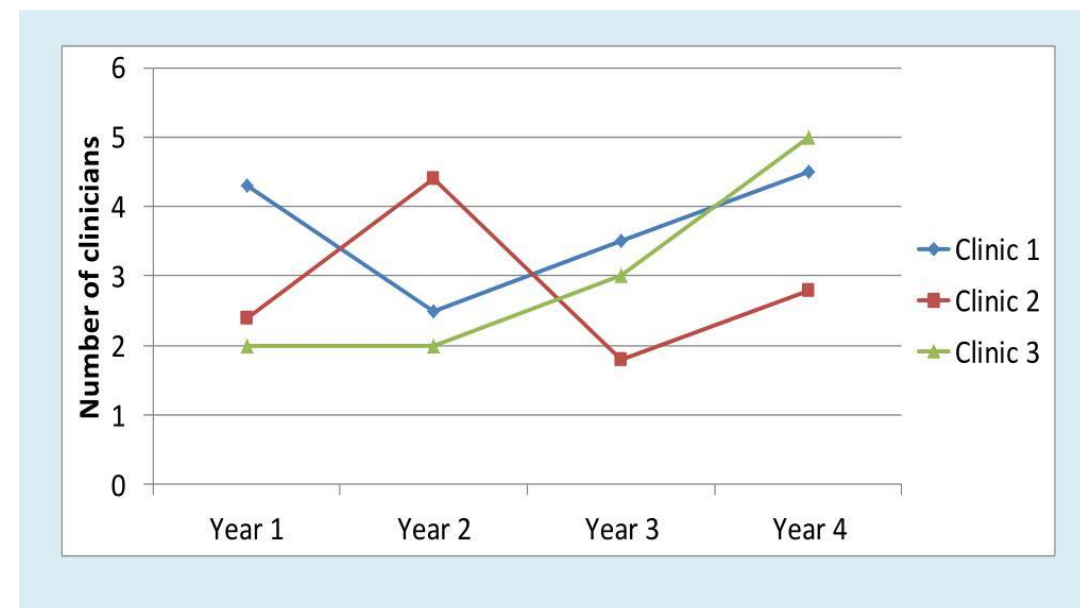
d- Pie Diagram

Pie charts are used to show the contribution of each item to the whole in which graphical description of data as slices of a pie and the total represents the complete pie. The values are commonly given as a percent or a proportion. It is used for **qualitative type of data**.



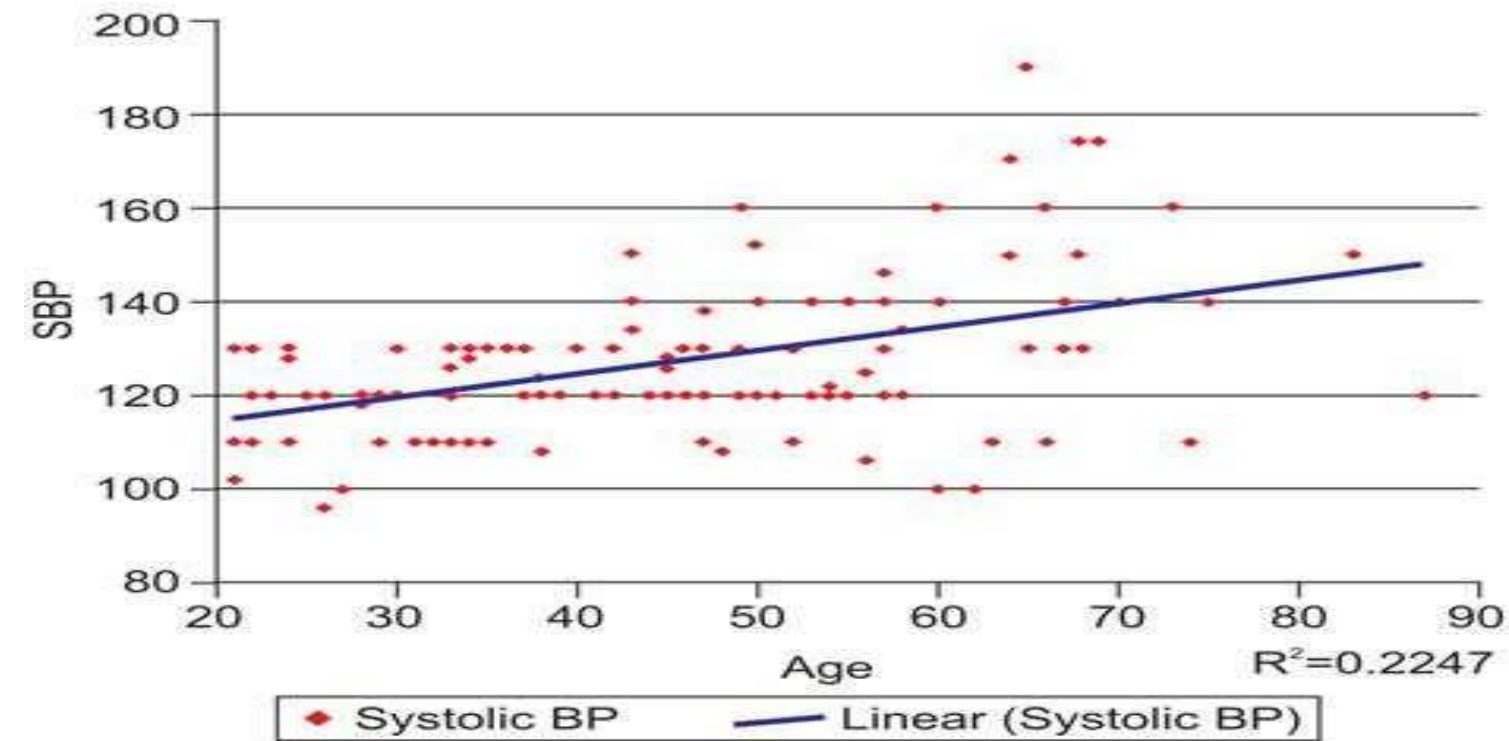
e-Line Graphs

Line graph is a graph that uses line segments to connect data points and shows changes in **data over time**. Each quantitative data value becomes a dot or point that is placed above the appropriate class values.



f-Scatter-plots diagram

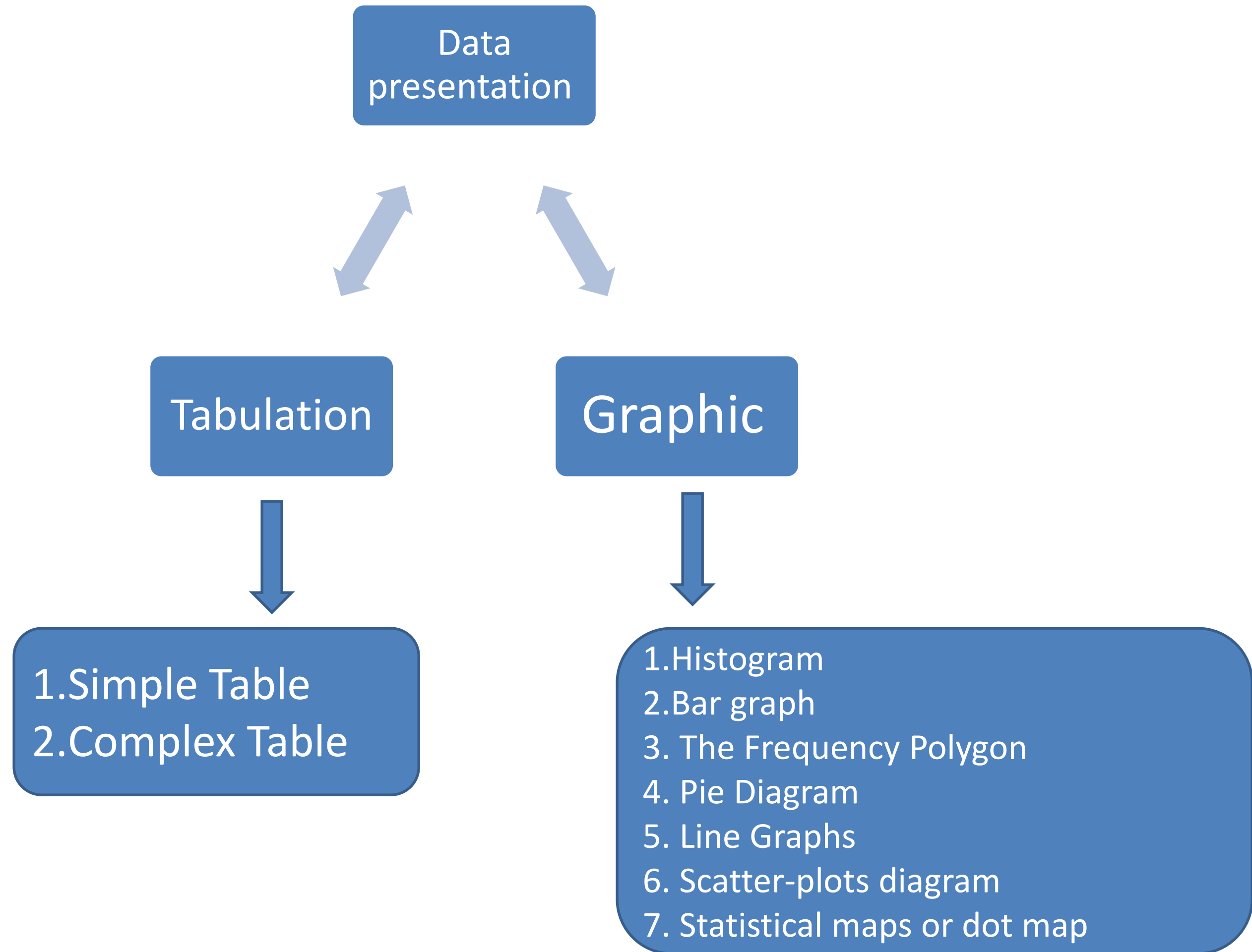
The scatter-plots visualize a relation (correlation) between two variables X and Y (both of them quantitative). If the dots cluster round a straight line, it shows evidence of a relationship of a linear nature. If there is no such cluster, it is probable that there is no relationship between the variables.



g-Statistical maps or dot map :

It is used when statistical data refers to geographic or regional distribution of a variable. The areas are shaded with different color and used to present data of varying size.





The background features abstract geometric shapes in various shades of blue and teal, primarily located in the corners and along the edges, creating a modern, layered effect.

THANK YOU